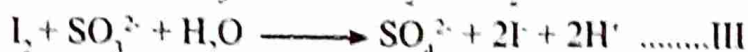


Experiment No. 5

Date: / /

Aim: To study the rate of reaction between potassium iodate (KIO_3) and sodium sulphite (Na_2SO_3) using starch solution as an indicator (clock reaction).

Theory: The reaction between KIO_3 and Na_2SO_3 indirectly involves the formation of iodide ions, which are oxidised in acidic medium by IO_3^- ions to iodine. The overall reaction proceeds in the following three steps.



The evolved iodine produces blue colour with the starch solution. As time required for appearance of blue colour is recorded to determine the rate of reaction, the method is called 'clock reaction'.

Apparatus: Four Conical flasks (250 mL), measuring cylinder (100 mL), stopwatch, pipette, etc.

Chemicals: 2M H_2SO_4 , 0.1M KIO_3 solution, 0.01M Na_2SO_3 solution, and starch solution.

Procedure:

1. Take four conical flasks (250ml). Clean with water and label them as A, B, C and D.
2. Take 10, 20, 30 and 40 mL of 0.1 M KIO_3 solution in flasks A, B, C and D respectively.
3. Add 10 mL of 2M H_2SO_4 to all flasks.
4. Add 80, 70, 60 and 50 mL of distilled water to the flasks A, B, C and D respectively to make the volume of 100 mL in each flask.
5. Add 5 mL of freshly prepared starch solution to each flask. (For prevention of oxidation)
6. Add 25 mL of 0.01 M freshly prepared Na_2SO_3 to flask A. (For prevention of oxidation) Shake and start stop watch and record the time required when blue colour just appears.
7. Repeat the procedure given in step 6 for solution in flask B, C and D.

Observation table :

Flasks	0.1 M KIO_3 /(mL)	2 M H_2SO_4 /(mL)	Water (mL)	Starch solution/(mL)	0.01M Na_2SO_3 /(mL)	Time required for appearance of blue colour
A	10	10	80	5	25	7.2...
B	20	10	70	5	25	6.0...
C	30	10	60	5	25	4.9...
D	40	10	50	5	25	3.8...

Result:

The rate of reaction between potassium iodate (KIO_3) and sodium sulphite (Na_2SO_3) using starch solution. As the time required for the appearance of blue colour

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

- What is the role of I^- ions when reacts with IO_3^- in acidic medium?
 - reducing agent
 - oxidising agent
 - precipitating agent
 - chelating agent
- The total amount of solution in each flask in above experiment is
 - 250 mL
 - 130 mL
 - 100 mL
 - 50 mL
- Why is it necessary to repeat the experiment in each case twice?
 - to take the average time
 - because there is no accuracy in appearance of blue colour.
 - the total amount of solution in each flask is not 100 cm³
 - there is temperature variation in water bath.
- Which one of the following acid is used to make the medium acidic in above experiment?
 - hydrochloric acid
 - perchloric acid
 - nitric acid
 - sulfuric acid

Short answer questions

1. Why fresh solution of sodium sulphite is used in this experiment?

Ans. Always use a freshly prepared solution of sodium sulphite because it is easily oxidized by air.

2. What will happen if we do not use a freshly prepared starch solution?

Ans. Starch solⁿ readily reacts with air, it forms a colloidal solⁿ which can interfere with blue colour formation of iodine-starch complex.

3. Mention the factors that affect the rate of reaction in above experiment?

Ans. The factor affecting rate of reaction is concentration of potassium iodate solution.

4. Out of the reactions (I) and (II) given below, which would be the rate determining reaction?

What is the molecularity of the rate determining step?



Ans. As reaction (I) is slow, it is rate determining reaction and its molecularity is 2 (bimolecular)

5. What is the role of iodate ion in above reaction?

Ans. In this protocol iodide ion is generated by reaction betⁿ iodate and bisulfite.

Remark and sign of teacher: